TITLE: **The trend of USD/TRY and EUR/TRY: moving averages, an exponential trend and exponential smoothing, checked on new months**

SUBJECT: Aim, method and results of version 2 of the `USD_Exp_MA_TrendAnalysis` repository

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Key finding. Both rates grew by about 2% a month, and an exponential trend fits them with R^2 of 0.960 (USD) and 0.950 (EUR); yet the trend leaves deviations that persist for years, and its forecast errors one month ahead are about eight times those of a random walk with drift. The trend of the exchange rate is stochastic, there is no monthly seasonality, and no smoothing method beats the drift by more than half a point at any horizon.

1. Data and design

Monthly averages of the CBRT indicative buying and selling rates (EVDS, TP.DK.USD.A/S.YTL and TP.DK.EUR.A/S.YTL), 2012-01 to 2026-09 (177 months), downloaded on 26 October 2026. As in version 1, the rate is the mean of the buying and selling rate. USD/TRY rose from 1.84 to 48.56 (a factor of 26.3), EUR/TRY from 2.38 to 55.96 (23.5); the mean monthly change was 1.86% and 1.79%. All series are $y_t = 100 \log(\text{rate}_t)$, so that changes are in per cent.

2. Trend: moving averages and the exponential trend (Figure 1)

Method. The 2×12 centred moving average, $\hat{T}_t = \frac{1}{24} y_{t-6} + \frac{1}{12} \sum_{j=-5}^{5} y_{t+j} + \frac{1}{24} y_{t+6}$, removes any fixed 12-month pattern and is centred on t; it is missing for the first and last six months. (The 3-month average of version 1, $\frac{1}{3}(y_t + y_{t-1} + y_{t-2})$, is trailing: it is centred on $t-1$, one month behind.) The exponential trend is $y_t = a + bt + u_t$ by OLS.

Finding. $\hat{b} = 2.09\%$ a month for USD and 2.00% for EUR. The residuals range from -43 to 50% (USD), their first autocorrelation is 0.98 and the Durbin–Watson statistic 0.033 (EUR: 0.029): the series stays above or below the trend for years. The ADF test does not reject a unit root in y_t around a linear trend ($p = 0.46$ and 0.52) and rejects it in Δy_t ($p < 0.001$).

Interpretation. A high R^2 here measures the common upward drift of y_t and t, not the quality of the trend; with a unit root the deviations from a fixed line do not die out.

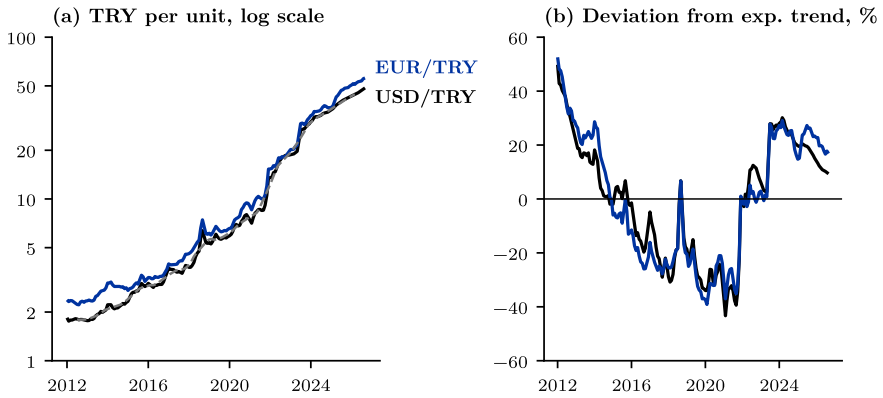


Figure 1. (a) USD/TRY (black) and EUR/TRY (blue), TRY per unit, log scale; dashed grey: 2×12 centred moving average of USD/TRY. (b) Deviation of y_t from the fitted exponential trend, per cent.

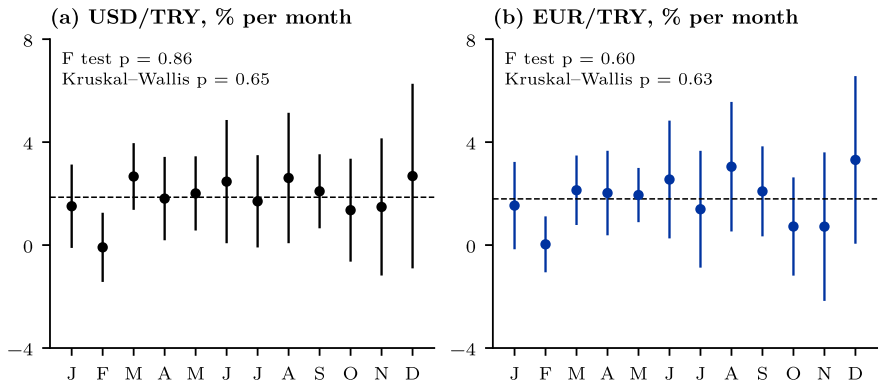


Figure 2. Mean monthly change of USD/TRY (a) and EUR/TRY (b) in each calendar month, with 95% intervals (±1.96 standard errors); dashed: mean over all months.

3. Seasonality (Figure 2)

Method. The monthly change Δy_t is regressed on a constant and 11 month dummies (F test), and compared across the 12 calendar months with the Kruskal–Wallis test.

Finding. Neither test finds a monthly pattern: $p = 0.86$ and 0.65 (USD), 0.60 and 0.63 (EUR). A seasonal component (Holt–Winters, seasonal decomposition) has nothing to estimate.

4. Forecasts on new months (Table 1, Figure 3)

Method. For each month from 2021-10 to 2026-09 (60 months), every method is fitted on the earlier months only and forecasts 1, 3 and 12 months ahead: random walk ($\hat{y}_{t+h} = y_t$), random walk with drift

Table 1. RMSE of forecasts on the 60 months from 2021-10 to 2026-09, per cent of the rate.

	USD/TRY			EUR/TRY		
Method	1 m	3 m	12 m	1 m	3 m	12 m
Random walk	5.2	12.6	39.2	5.3	12.4	37.0
Random walk with drift	4.5	10.1	25.2	4.6	9.9	22.3
Exponential trend	36.0	38.1	47.3	37.9	40.1	49.9
Simple exponential smoothing	5.2	12.6	39.2	5.3	12.4	37.0
Holt	5.5	13.4	42.4	4.7	10.2	23.3
Damped Holt	5.2	11.6	28.2	4.6	9.9	21.9
Holt–Winters (additive seasonal)	5.8	13.8	42.7	4.9	10.4	21.8

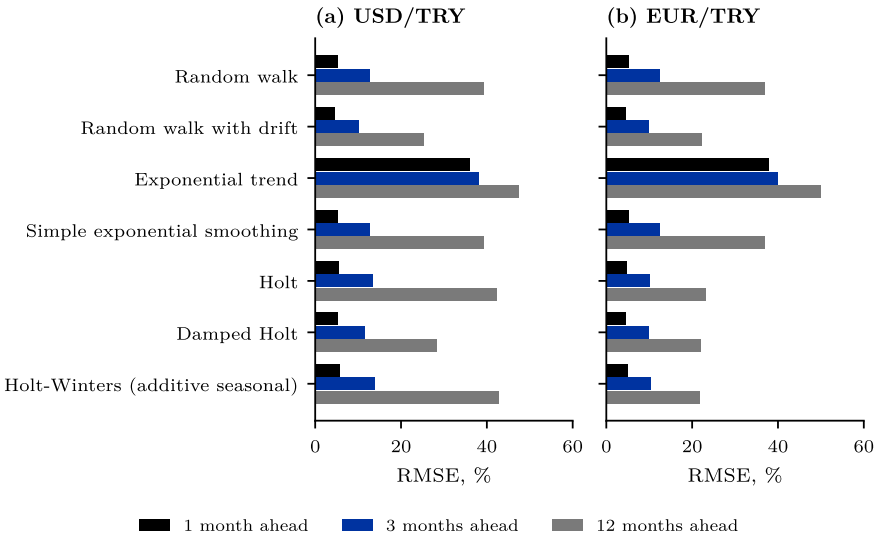


Figure 3. RMSE of the forecasts on new months, by method and horizon.

($\hat{y}_{t+h} = y_t + h\bar{\Delta}y$), exponential trend, simple exponential smoothing, Holt, damped Holt and Holt–Winters (additive, 12 months). Accuracy is the root mean squared error (RMSE) in per cent.

Finding. The exponential trend has the largest errors at every horizon: 36.0% one month ahead for USD against 4.5% for the drift. Simple exponential smoothing equals the random walk (the fitted smoothing weight is 1). For USD the drift is the most accurate at every horizon (25.2% at 12 months); for EUR the drift, damped Holt and Holt–Winters are within half a point of one another at 12 months (22.3%, 21.9% and 21.8%).

5. Conclusions and limitations

1. The trend of USD/TRY and EUR/TRY is stochastic: a fixed exponential trend fits the past closely but forecasts poorly.
2. There is no monthly seasonality in the changes, so seasonal methods add nothing.
3. The random walk with drift is a hard benchmark: for USD it is the most accurate at every horizon, for EUR damped Holt and Holt–Winters are within half a point of it.
4. Limitations: one evaluation window (2021–2026) that includes large depreciations; RMSE only; no test of the differences between methods.

Scope and verification note. Every number in this note is written by `make_paper.py` from the CSV tables of the analysis. `tests/run_tests.py` checks the moving average, the trend regression, the seasonality tests and the forecasting scheme against independent computations, and reproduces version 1's output ($R^2 = 0.945$ on its own data).

References

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